AR TACTICAL HELMET

The Problem

- Soldiers use multiple devices for maps, compasses, and data, causing distractions.
- Each glance at a device results in a 1-2 second loss of situational awareness.
- This momentary distraction can be life-threatening in combat situations.
- Oxygen level deplation at high altitudes and no Heart-rate measurement

The Project:

- Provides live navigation and compass directions.
- Displays rear camera view and thermal data on the visor.
- Ensures hands-free operation for improved situational awareness.
- Displays SpO2 & Heart-rate levels
- Inter-connectivity between Helmets
- Live map updates through command center

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TOTAL COST

500,000 PKR
With Xreal X1 Pro AR Glasses

300,000 PKR
Without Xreal X1 Pro AR Glasses

FYP Cost Breakdown		
#	Component	Cost (PKR)
1	Raspberry Pi 5 (8GB)	59,000
2	ZT TFT SPI Module GC9A01	4,000
3	HUD Reflector Lens	400
4	Raspberry Pi Camera Module 3	17,000
5	Original IMX291 USB Camera Module	20,000
6	Digital Infrared Temp Sensor MLX90614	2,000
7	IMU Head Tracking	7,300
8	Helmet 3D Printed Prototype	TBA
9	Power Supply	500
10	Cabling and Wires	TBA
11	ESP32	1,500
12	Sony ECX339A OLED Panel	22,000
13	DLPDLCR2000EVM Pico Projector	35,000
14	Mini OLED Transparent Display 2x	30,000
15	Xreal X1 Pro (Back-up Option)	200,000
16	Operations & Logistics	70,000
17	Filament	30,000
18	Pulse Sensor Module	400
19	SpO2 Sensor	450
	TOTAL	**299,550**

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